Thus we see that the projected version of the symplectic form in the ambient space is conserved by the differential equation system (4.7)–(4.9). This is what is meant by saying that the constrained system is analogous to a Hamiltonian system. We may also think of its flow map as being a symplectic map of the co-tangent bundle.

4.3.2 Numerical Methods with Holonomic Constraints

In what follows, we discuss the development of constrained integration methods, with a focus on the symplectic structure. Although we do this in a different way here, our presentation is based on the article [229].

From a mathematical point of view, the natural concept of a numerical method for the holonomic system (4.7)–(4.9) is that of a mapping from $T^*\mathcal{M}$ to itself which approximates the dynamical evolution on a step of length h.

Actually, many methods in common use for molecular dynamics cannot be seen as maps of $T^*\mathcal{M}$ since the hidden constraint is allowed to be violated. A simple example of such a method is the constrained Symplectic Euler-like method

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1}, \tag{4.14}$$

$$\boldsymbol{p}_{n+1} = \boldsymbol{p}_n - h\boldsymbol{F}_n - h\boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_n, \tag{4.15}$$

$$\boldsymbol{0} = \boldsymbol{g}(\boldsymbol{q}_{n+1}). \tag{4.16}$$

Here $\boldsymbol{\lambda}_n$ is a vector which needs to be computed to define the projection onto $\mathcal{M}$. We would typically assume that $\boldsymbol{q}_0$ lies on $\mathcal{M}$ so that (4.14)–(4.16) defines a map on $\mathcal{M}$. The hidden constraint $\boldsymbol{g}'(\boldsymbol{q}_0)\boldsymbol{M}^{-1}\boldsymbol{p}_0 = 0$ will be violated, but one hopes it remains approximately satisfied. The justification for the latter assumption is that if we expand $\boldsymbol{g}$ in a Taylor series, we may write

$$\boldsymbol{g}(\boldsymbol{q}_n) \approx \boldsymbol{g}(\boldsymbol{q}_{n+1}) + \boldsymbol{g}'(\boldsymbol{q}_{n+1})(\boldsymbol{q}_n - \boldsymbol{q}_{n+1}).$$

Then since we maintain $\boldsymbol{g}(\boldsymbol{q}_n) = 0$ for all n, we have

$$\boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1} \approx 0.$$

To solve the Eqs. (4.14)–(4.16), insert the second in the first and the first then in the third to obtain the simplified equation

$$\boldsymbol{g}(\boldsymbol{Q}_n - \boldsymbol{M}^{-1}\boldsymbol{G}_n^T\boldsymbol{\Lambda}) = 0, \tag{4.17}$$

where

$$\begin{aligned}\boldsymbol{Q}_n &= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_n - h^2\boldsymbol{M}^{-1}\boldsymbol{F}_n,\\ \boldsymbol{G}_n &= \boldsymbol{g}'(\boldsymbol{q}_n),\\ \boldsymbol{\Lambda} &= h^2\boldsymbol{\lambda}_n.\end{aligned}$$